

Nate Cirino

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Graduating May 2027 | Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Algorithms & Data Structures, Computer Systems, Computer Networks, Object-Oriented Design, Foundations of Software Engineering, Digital Design & Computer Architecture, Systems Security, Cybersecurity

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-labels, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 heterogeneous supplier integrations behind one access layer
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Python/Flask RAG pipeline over Bedrock Knowledge Bases, grounding LLM output in indexed domain data
- Evolved retrieval stack from LangChain to Vespa to zero-ops Bedrock KB, improving relevance and cutting overhead
- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on architecture and agentic-workflow direction

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts on high-throughput event streams
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing GenAI workflow
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and batch data pipelines in Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

Sentinel | *FastAPI, Claude, pgvector, RAG, OpenTelemetry, Next.js, Docker*

Apr. 2026 – Present

- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Built an AI SRE agent that reasons and acts over Prometheus, Tempo, and Loki telemetry via LLM tool-use
- Grounded diagnoses in cited retrieved context; human-approved remediation, live in SculptAI production

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits

SKILLS & HONORS

Languages/Tools: Python, Java, Rust, C++, C, TypeScript, JavaScript, Bash, PostgreSQL, GraphQL, Linux, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, Buildkite, Groovy

LLM / RAG & Agents: Bedrock KB, pgvector, Vespa, LangChain, agentic tool-use, Cursor, Claude Code, Copilot

Frameworks & Libraries: PyTorch, TensorFlow, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor